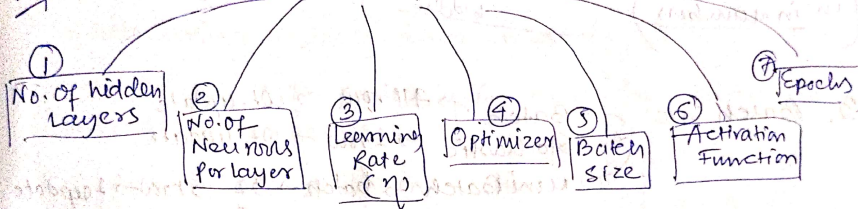


How to Improve a NN

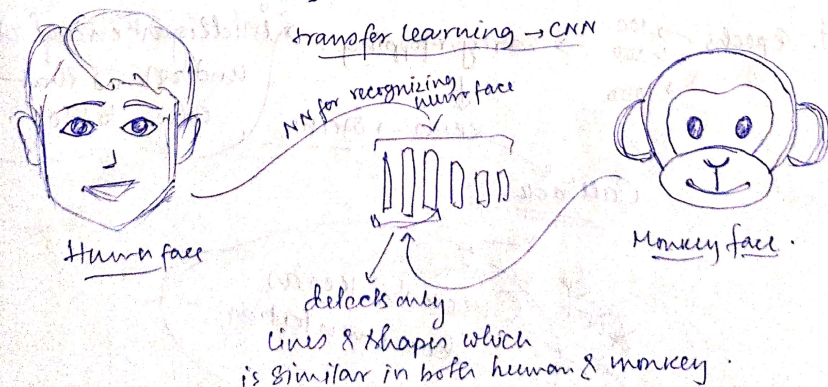
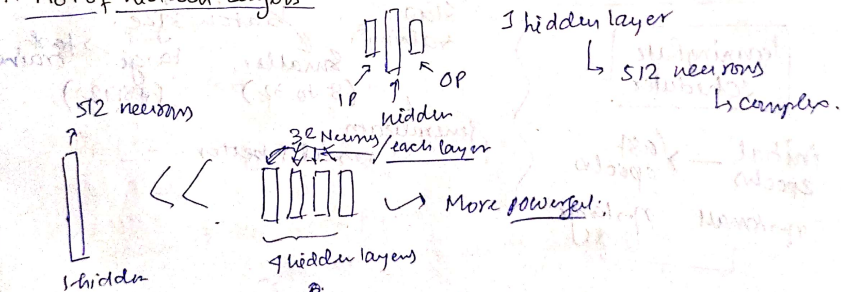
1. fine tuning NN hyperparameters.
 - # hidden layers
 - # neurons/layer
 - $\eta \rightarrow \text{Go}$
 - Optimizer
 - Batch size
 - Activation function
 - Epochs.
2. By Solving problems:
 - vanishing/Exploding gradient.
 - Not enough data.
 - slow training
 - Overfitting.

22/08/23

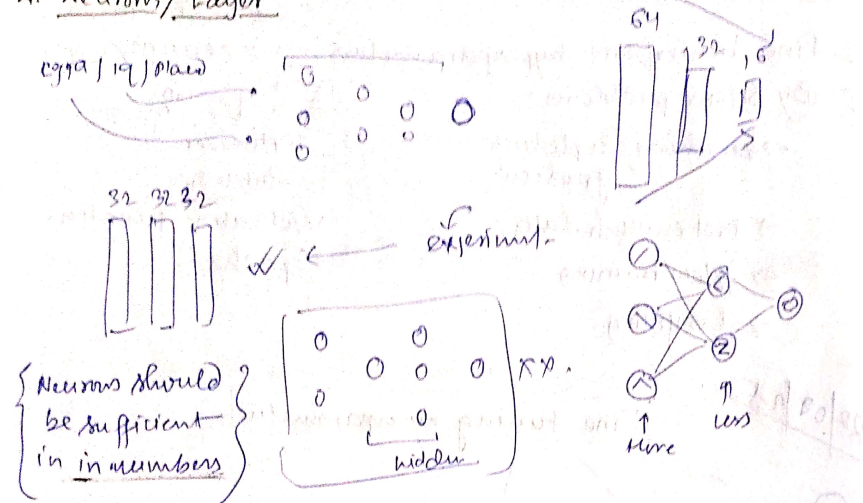
Fine tuning hyperparameters



1. No. of hidden layers

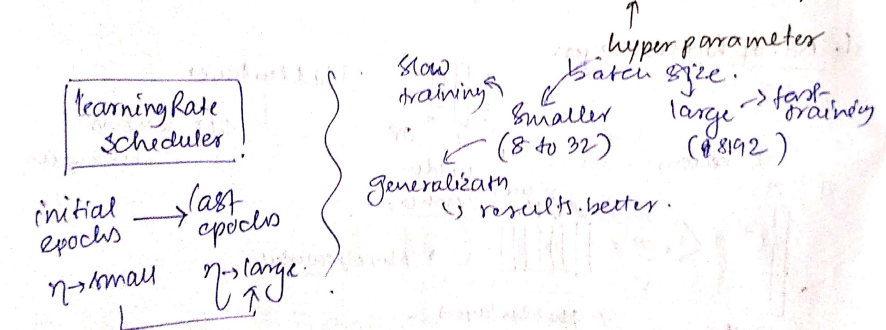


2. Neurons/layer



2. Batch size

- Batch → All row → cost. update
- Stochastic → 1 row → wt. update
- mini Batch → batch → 32 → row → 1 update



4. Epochs
 - 100
 - 500
 - 1000
 - Early Stopping
 - Keras → stable
 - intelligent enough to understand that when to stop.

Callback.

Check in Keras documentation